National Institute of Technology Patna
Department of Electronics and Communication Engineering
Subject Code: EC031102/EC031202/EC031301 (Digital Electronics)
MID-SEM EXAMINATION
DOE: 16/10/2026 (FN, 10:00 AM to 12:00 PM)

Semester: 3rd **MM: 30**

Course: B.Tech- ECE/B.Tech-M.Tech-DD-ECE-MV/EE-VLSI **Time: 2 Hrs.**

Note: Answer all questions. **[3x10=30 Marks]**

1. **(a)** Why are digital circuits called switching circuits? **[CO-1] [2M]**
   (b) Define Idempotent Law. **[CO-1] [2M]**
   (C) Show AB'C+B+BD'+ABD'+A'C = B+C **[CO-2] [2M]**
   d) Design XOR gate using 4x1 MUX. **[CO-2] [2M]**
   (e) Draw the block diagram of a full subtractor using two half adders. **[CO-2] [2M]**

2. **(a)** Design and implement an 8×1 multiplexer using 2×1 multiplexers. Draw the logic diagram. **[CO-1, CO-2,CO-4] [4M]**
   (b) Design JK using D Flip Flop. **[CO-4][3M]**
   (c) Define race condition and race-around condition. Draw the circuit used to eliminate the race-around condition. **[CO-1, CO-2] [3M]**

3. **(a)** Using the Quine- Mccluskey minimization method. Obtain the minimal expression for F=(A,B,C,D) = m ∑(0,1,5,7,8,10,14,15) **[CO-2] [4M]**

   (b) What is a magnitude comparator? Write the Boolean expressions for a 4-bit magnitude comparator. **[CO-1,CO-2] [3M]**
   (c) Simplify the Boolean function $F(A,B,C,D) = \sum m(0,1,2,3,5,8,9,10,12,13)$ using a Karnaugh map (K-map), and implement the minimized expression using universal logic gates (NAND Gate) **[CO-2] [3M]**